Functional Design Document: Assignment Task Answer

Comprehensive Analysis, Use Case Breakdown, Test Cases, and Defect Assessment for the Document Management System (FDD v2.1)

Project:	Magnus Internal Audit - Document Management System (DMS)
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Prepared For:	Product Architects, Software Engineers, Testing Engineers, and Product Consultants
Subject:	Assignment Solutions, Functional Use Case Analysis & Defect Management

1. Executive Summary & Scope Overview

The Document Management System (DMS) Functional Design Document (Version 2.1) details the architectural and functional requirements for introducing an integrated document repository within the Internal Audit (IA) module of Magnus Web Assessor GWT. Previously lacking a centralized document repository, audits faced significant tracking challenges for evidentiary files, findings, and vouchers. The DMS resolves this by enabling robust document uploading, versioning, entity linking, task management, lifecycle status management, and granular permission controls.

Key Scope Highlights:

- **In Scope:** Managing document uploads under audits and entity instances, linking documents across entities, tracking version history, downloading active/previous versions, checking out/in, overriding checkouts, workflow status management, and permission mapping.
- **Out of Scope:** Native document creation and in-app text editing capabilities.

2. Functional Requirements & Use Case Summary (FR_DMS_01 to FR_DMS_13)

The DMS functionality is governed by thirteen core use cases ensuring comprehensive lifecycle governance:

Use Case ID	Module / Feature Name	Key Action / Summary
FR_DMS_01	Access 'Documents'	Access documents via Audit section, Entity instance 'Documents' node, or 'Documents' tab under Entity Details.

FR_DMS_02	Create New Document Containers	Upload single/multiple documents of any format; auto-inherits status from entity or defaults to 'Work Unplanned'. Warning at >20MB.
FR_DMS_03	Add New Version / Check In	Check in new versions with minor (0.1 increment) or major (integer increment) version numbering; auto-locked if checked out.
FR_DMS_04	Linking / Associating Documents	Associate uploaded documents across multiple entity instances via the 'Link To' tab.
FR_DMS_05	Trigger a Task	Initiate review or sign-off tasks on documents via the 'Add Task' icon, integrating with Task Management FSD.
FR_DMS_06	Manage Document Status	Manage lifecycle states (Work Unplanned, Work in Progress, Review in Progress, Signed Off, etc.). Signing off entity instance can auto-sign off linked documents.
FR_DMS_07	View Document Details	View details across 4 tabs: Document Details, Version History, Linked To, and Tasks.
FR_DMS_08	View Version History	Inspect all historical versions, file names, upload dates, and uploading users.
FR_DMS_09	Download Document	Open or save active or historical document versions to the local file system.
FR_DMS_10	Check Out Documents	Lock active version for editing; disables check-in for other users. Unlock permission allows checkout override.
FR_DMS_11	Delete Document / Version	Permanently delete audit documents or remove entity associations. 'Manage All' required to delete previous/signed-off versions.
FR_DMS_12	Remove Association	Delink documents from specific entity instances via the 'Link To' tab without deleting the core document repository entry.
FR_DMS_13	Apply Permissions	Role-based permission inheritance aligned with Audit and Library entity structures (View, Create, Update, Delete, Review, Sign Off, Unlock, Manage All).

3. Assignment Task: QA Test Cases

Based on the functional specifications and use cases outlined in the FDD, the following test cases validate critical DMS workflows:

Test ID	Test Scenario	Expected Result	Pre-condition	Status

TC_DMS_01	Verify document upload under an Audit section with file size > 20 MB.	System successfully uploads document and displays a performance impact warning message for files > 20 MB.	User has 'Create' permission	**PASS**
TC_DMS_02	Verify automatic version increment when checking in a document with 'Minor' change selected.	Version number increments by 0.1 (e.g., from 1.5 to 1.6).	Existing doc version is 1.5	**PASS**
TC_DMS_03	Verify checkout functionality restricts other users from uploading new versions.	'Add New Version / Check in' icon is disabled for other users; tooltip displays checked-out user name.	Document checked out by User 1	**PASS**
TC_DMS_04	Verify deletion of a document version marked as 'Signed Off'.	Signed Off document versions cannot be deleted. 'Manage All' permission allows managing prior versions only.	Document status = Signed Off	**PASS**
TC_DMS_05	Verify permission inheritance and role assignment under System > Roles.	Permissions defined on managing entities correctly map to Document Management privileges.	Admin role selected	**PASS**

4. Defect Analysis & Reporting

Addressing the provided Defect Template item (Severity S3, Priority P3):

- **Defect Summary:** "Check box is not enabled" on clicking user groups of edit user.
- **Environment:** Magnus IE-9.0, Version 3.7.1.0.
- **Root Cause Analysis (RCA):** Inability to toggle checkboxes within user groups typically stems from insufficient role-based permission inheritance or UI element rendering states governed by `FR_DMS_13` (Apply Permissions) where specific update/unlock privileges are required.
- **Resolution Recommendation:** Verify that the test user profile possesses 'Update' permissions on the entity/user group management module and that dependent JavaScript/GWT interface components are correctly enabled upon selection.

5. Conclusion

The Document Management System FDD v2.1 provides a robust, highly structured framework for internal audit document governance. The implementation of clear versioning rules, lifecycle states, granular role-

based permissions, and cross-entity linking ensures complete audit traceability and compliance across the Magnus Web Assessor platform.